

Inverse of 2x2 matrices

Key Points

The inverse of M is M^{-1} such that
 $MM^{-1} = M^{-1}M = I$

$$\text{For } M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$M^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

In general $M^{-1} = \frac{1}{|M|} \text{Adj}(M)$

Where $\text{Adj}(M)$ is the adjugate matrix.

$$(AB)^{-1} = B^{-1}A^{-1}$$

Basic Skills

Q1. Find the following determinants

a) $\begin{vmatrix} 3 & 1 \\ 5 & -2 \end{vmatrix}$ b) $\begin{vmatrix} 1 & 3 \\ 0 & 5 \end{vmatrix}$ c) $\begin{vmatrix} 5 & 1 \\ 10 & 2 \end{vmatrix}$

Q2. State the adjugate of the following matrices

a) $\begin{pmatrix} 5 & 3 \\ 1 & 2 \end{pmatrix}$ b) $\begin{pmatrix} 2 & 2 \\ -1 & 1 \end{pmatrix}$ c) $\begin{pmatrix} 4 & -1 \\ 0 & 2 \end{pmatrix}$

Q3. Find the inverse of the following matrices

a) $\begin{pmatrix} 3 & 0 \\ 2 & 1 \end{pmatrix}$ b) $\begin{pmatrix} 5 & 1 \\ 2 & 1 \end{pmatrix}$ c) $\begin{pmatrix} 5 & 4 \\ 2 & -3 \end{pmatrix}$

Q4. Explain when a matrix does not have an inverse

Competency Skills

Q5. Prove that $M^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$

Q6. The 2D Euclidean plane is transformed by $M = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$

a) Find the inverse transform M^{-1}

b) Find the point which maps to the point (3,4) by using the inverse transform

Q7. Explain why $(AB)^{-1} = B^{-1}A^{-1}$

Q8. The matrix $\begin{pmatrix} k & 2k \\ 3 & k \end{pmatrix}$ has no inverse find all values of k

Q9. Given $A = \begin{pmatrix} 3 & 5 \\ 1 & 3 \end{pmatrix}$ and $AB = \begin{pmatrix} 5 & 3 \\ 0 & 2 \end{pmatrix}$ find the matrix B

Advanced Skills

Q10. The unit square is mapped onto the quadrilateral (0,0),(5,4),(2,3),(8,10) is such a transformation possible by matrix multiplication and if so find the inverse transform

Q11. A matrix is called a self-inverse if $M^{-1} = M$ determine the conditions for a 2x2 matrix to be a self-inverse and give an example of a standard transform fitting these conditions

Q12. Given $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is uninvertible prove that $M = (a+d)^{n-1} \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

Q13. A student says "It is true for all matrices that $AA^{-1} = I$ " comment on there statement

Q14. Given that $ABCD = I$ show algebraically that $B^{-1} = CDA$

Extension

Q15. When a matrix is uninvertible what does it mean about the transform the matrix represents